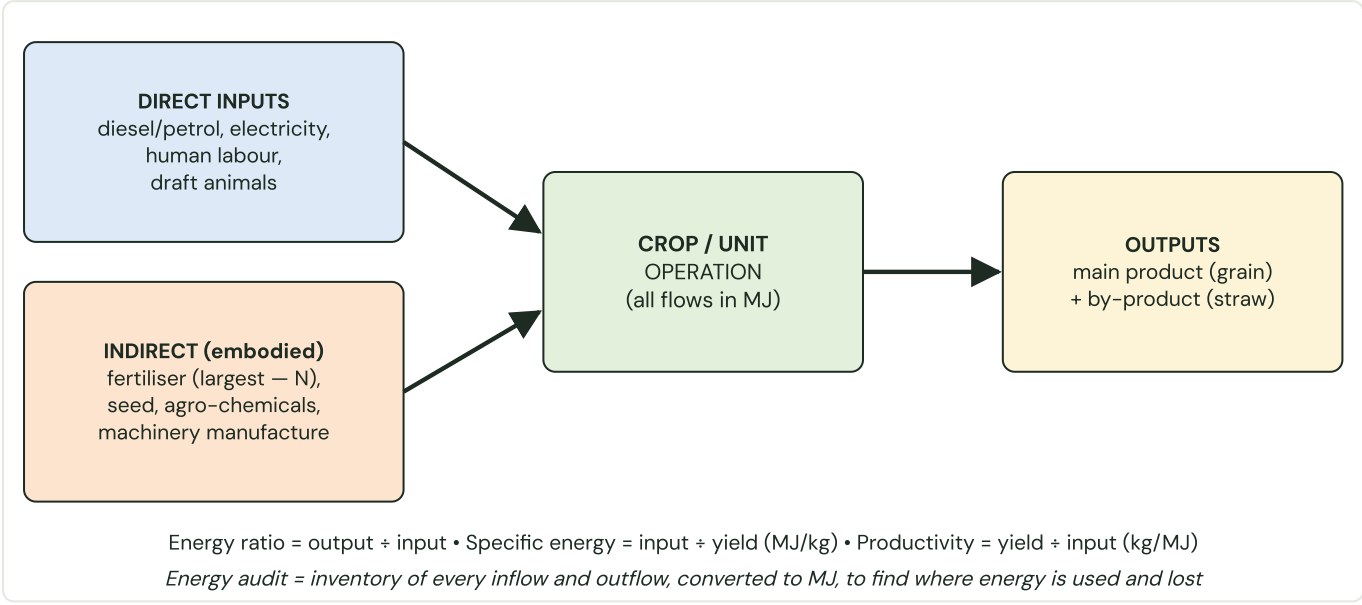
UNIT 7 · TOPIC 12

Energy Inflow and Outflow — Energy Audit, Accounting and Analysis

Syllabus topic: Inflow and outflow of energy in unit agricultural operations; energy audit, accounting and analysis



Energy inflow and outflow of a crop or unit operation (new figure; the source chart showed two uninformative 100% bars)

Step	What it does
ENERGY AUDIT	A systematic inventory of ALL energy INFLOWS and OUTFLOWS of a unit operation, crop or farm, converted to MJ, to locate where energy is used and lost
ENERGY ACCOUNTING	Tabulates inputs by source (direct/indirect, commercial/non-commercial, renewable/non-renewable) and outputs (main product + by-product)
ENERGY ANALYSIS	Computes the indices (energy ratio, specific energy, productivity — Topic 11) and benchmarks operations to target improvements

Carrying out an energy audit

1. List every input of the operation or crop: diesel, electricity, labour, animal power, fertiliser, seed, chemicals, machinery use.
2. Convert each to MJ with standard energy equivalents (Topic 11).
3. List the outputs — main product and by-product — and convert them to MJ.
4. Compute the energy ratio, specific energy and energy productivity.
5. Identify the largest inputs and losses, and target them.

Typical improvements

- Cut the fertiliser (indirect) share.
- Replace diesel pumping with solar (Topic 4b).
- Reduced/zero tillage.

- Better machinery matching (Unit 3, Topic 10a).

Purpose: raise the energy ratio and lower specific energy while sustaining yield.

Practice MCQs

Q139. An energy AUDIT is:

(a) A financial statement

(b) A systematic inventory of all energy inflows and outflows converted to MJ

(c) A soil test

(d) A yield forecast

Q140. A way to improve the energy ratio of a cropping system is to:

(a) Increase nitrogen use

(b) Replace diesel pumping with solar and reduce tillage

(c) Add more machinery

(d) Irrigate more often

Answers

Q139 — (b) A systematic inventory of all energy inflows and outflows converted to MJ

Q140 — (b) Replace diesel pumping with solar and reduce tillage

Sources: Rai — Non-Conventional Energy Sources; Mittal — Energy in Agriculture; MNRE Biogas/SATAT programmes & PIB (as given in the source notes, July 2026); IS 16087:2016 (CBG specification); standard bioenergy texts; NPTEL bioenergy & energy-management lectures.